Ex no. 2A Study of Different types of Network Cables

10/8/26

Aim: Understand that different types of network cables.

Different types of cables used in networking are:

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial cable
4. Fiber Optic cable.

Cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
UTP	Category 3	10 bps	Advantages: • cheaper in cost • easy to install as they have a smaller overall diameter Disadvantages: • More prone to EMI (Electro-magnetic interference) and noise	10 Base - T Ethernet
	Category 5	upto 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	upto 1 Gbps		Fast Ethernet, Gigabyte ethernet
STP	Category 6, 6a	10 Gbps	Advantages: • Shielded • Faster than UTP • less susceptible to noise and interference Disadvantages: • Expensive • Greater installation effort	Gigabit Ethernet 10G Ethernet (55m) widely used in data centre
	Category 7	10 Gbps		Gigabit Ethernet 10G Ethernet (room)

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
Coaxial	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: • High bandwidth • Immune to interference • Low loss bandwidth • Versatile Disadvantages: • Limited distance • Cost • Size is bulky	Speed of signal is 500 Television network, high speed internet connections
Fibre optics-cable	Single mode, Multi mode	100 Gbps	Advantages: • High speed • High bandwidth • High security • Long distance Disadvantages: • Expensive • Require skilled installers	Maximum distance of fibre optics cable is around is 100m.

Result:-

Hence different types of network cables were studied and understood successfully.

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